



[WAREHOUSE & 3PL MANAGEMENT SYSTEM]

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Abstract

- Warehouse & 3PL Management System** is a centralized platform for managing **inventory, orders, warehouses, and shipments** using **React, Node.js, Express, MySQL, FastAPI, MongoDB, Kafka, and Qdrant**.
- Outcome:** Improves **warehouse efficiency, order fulfillment, and shipment tracking**.

Problem Statement & Objectives

- Problem:** [Traditional warehouse and 3PL operations use disconnected systems, making it difficult to manage inventory, orders, warehouses, and shipments efficiently.]
- Objective 1:** [Centralize inventory, order, warehouse, and shipment management.]
- Objective 2:** [Provide efficient order processing and real-time shipment tracking.]
- Objective 3:** [Enable role-based access and maintain accurate operational records.]
- Scope:** [For **administrators, warehouse staff, and logistics providers**, covering inventory, orders, warehouses, shipments, and activity tracking.]

Existing vs Proposed System

Feature	Existing System	Proposed System
[Inventory]	[Manual tracking]	[Centralized inventory]
[Orders]	[Separate management]	[Easy order management]
[Shipments]	[Limited tracking]	[Shipment status tracking]
[Warehouse]	[Manual operations]	[Digital warehouse management]
[Limitation / Gap]	[Data is scattered]	[One integrated system]

Feasibility Study

- Technical:** [Uses available tools like React, Node.js, MySQL, and VS Code.]
- Economic:** [Uses open-source technologies with low development cost.]
- Operational:** [Simple interface with easy maintenance and usage.]
- Schedule:** [Completion within the semester timeline]

Results & Testing

- [Successful inventory, order, warehouse, and shipment management.]
- [CRUD operations, input validation, error handling, efficient API responses, and optimized database operations were successfully implemented and tested.]

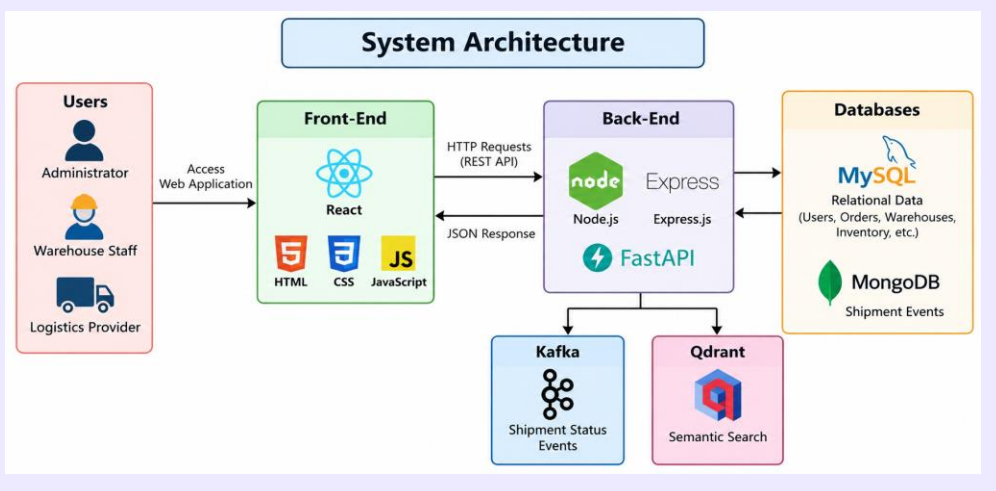
Conclusion

- [Successfully developed a centralized system for managing warehouse, inventory, orders, and shipments.]
- [The system meets the objectives by providing efficient management, validation, tracking, and role-based operations.]

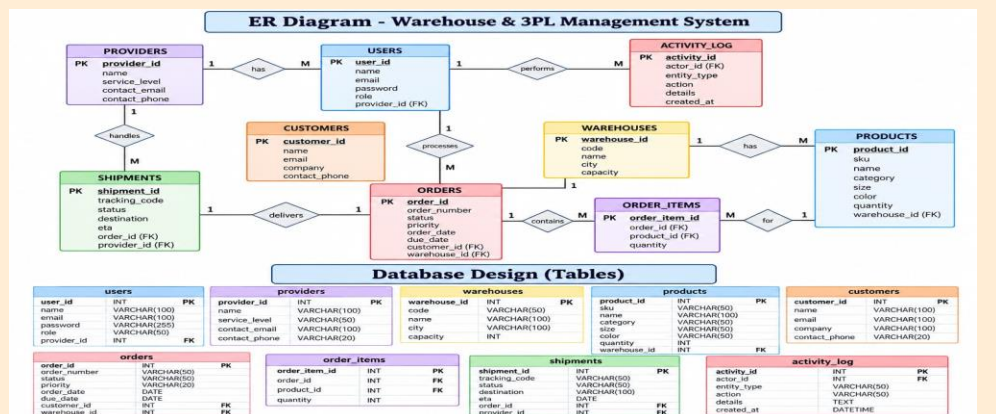
Tools & Technologies

- Front-End: [React / HTML-CSS-JS]
- Back-End: [Node.js / Express.js / FastAPI] + REST APIs
- Database: [MySQL / MongoDB]
- Tools: [Git & GitHub, Postman, VS Code, Docker]
- Others: [Kafka, Qdrant]

System Architecture



ER Diagram & Database Design



Implementation Screenshots

